

## PROJECT & INFRASTRUCTURE REPORT

# Bullion Infrastructure

A real, CI/CD-deployed multi-cloud Kubernetes platform running AWS EKS and GCP GKE as **one active-active cluster** — joined by an Istio service mesh, with live traffic load-balanced across both clouds.

EKS + GKE

Istio multi-primary

Active-active LB

GitHub Actions + OIDC

Terraform IaC

✓ Active-active cross-cloud mesh LIVE & verified — traffic balances EKS ⇄ GKE (12 AWS / 8 GCP)

Repo: Olisaemeka111/bullion-infrastructure · AWS acct 730767193290 · GCP project-f38bfcd6 · Report 2026-06-23

AWS / EKS

GCP / GKE

Service mesh

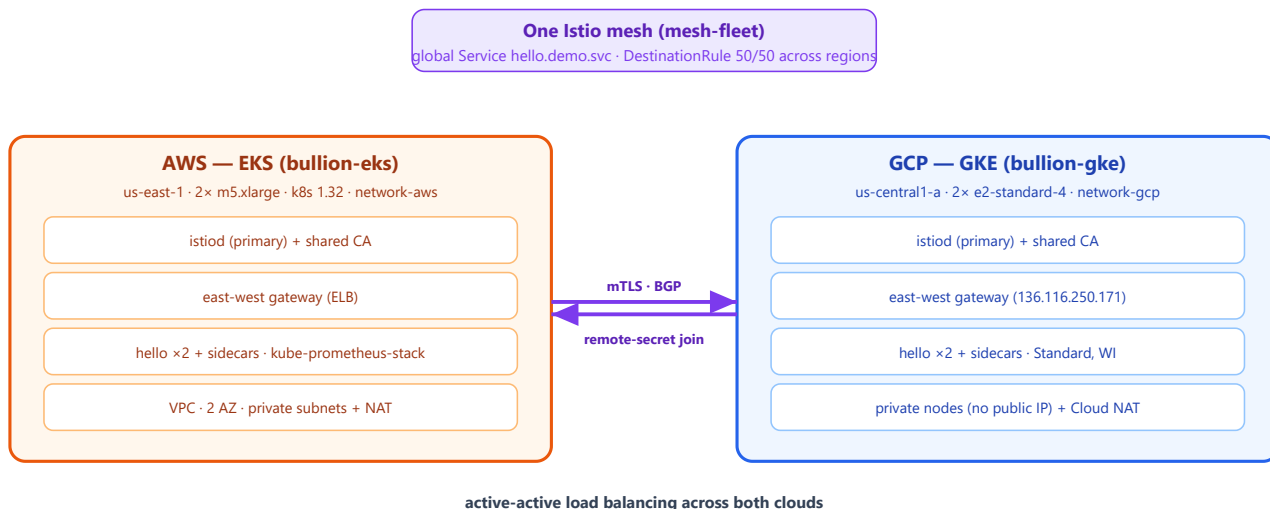
CI/CD

Observability

Pending

## 1 Goal & architecture

Run clusters in more than one cloud and have them behave as **one active-active platform** — not primary + cold backup. A workload deployed to both clouds is one global service; requests load-balance across both at once; if a cloud fails, traffic auto-redistributes to the survivor. All infrastructure-as-code + CI/CD, keyless auth, gated approvals.



## 2 Cloud inventory (final state)

### AWS · US-EAST-1 EKS — bullion-eks

|         |                                                                                   |
|---------|-----------------------------------------------------------------------------------|
| Cluster | EKS, Kubernetes <b>1.32</b>                                                       |
| Nodes   | 2 × <b>m5.xlarge</b> (4 vCPU/16GB)                                                |
| Network | VPC, 2 AZ, private subnets + 1 NAT, public subnets                                |
| Access  | EKS access entries (CI role + Olisa = admin); node SG open for istiod 15017/15012 |
| State   | S3 <code>fleet-mini-tfstate-59998ac4</code> , native locking                      |

### GCP · US-CENTRAL1-A GKE — bullion-gke

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Cluster     | GKE <b>Standard</b> (zonal), Workload Identity                          |
| Nodes       | 2 × <b>e2-standard-4</b> (4 vCPU/16GB)                                  |
| Network     | VPC-native; <b>private nodes</b> + Cloud NAT egress                     |
| Why private | free-trial IP quota (4) — keeps node IPs off the count for the mesh LBs |
| Project     | <code>project-f38bfcd6-70ff-4a8d-9cd</code>                             |

**Azure** — disabled. The account has no subscription ( `NoAzurePlanFound` ); the azurerm provider + AKS module are commented out until one exists. Re-enabling is a documented toggle, no redesign.

### 3 CI/CD & security

All deploys run through **GitHub Actions** with **keyless OIDC** auth and a gated `production` environment (required reviewer). No static cloud keys.

| WORKFLOW                  | PURPOSE                                                             | TRIGGER                 |
|---------------------------|---------------------------------------------------------------------|-------------------------|
| <code>ci.yml</code>       | 80 stdlib tests + <code>terraform fmt/validate</code> (no creds)    | every push / PR         |
| <code>deploy.yml</code>   | Terraform: EKS, GKE, cross-cloud VPN — plan → gated apply → destroy | PR / push / dispatch    |
| <code>platform.yml</code> | Istio multi-primary join + global app + observability               | dispatch / mesh changes |

#### Keyless cloud auth

**AWS:** GitHub OIDC → IAM role `fleet-mini-gha`. **GCP:** OIDC → Workload Identity Federation → SA `fleet-mini-gha`. No long-lived keys.

#### State & secrets

S3 **native state locking** (no DynamoDB). Cloud auth is keyless OIDC; only the non-sensitive OIDC identifiers are stored, as **GitHub Actions secrets/variables** (set via `iac/generate-credentials.sh` using the `gh` CLI).

### 4 Service mesh — final state

The two clusters form **one mesh** ( `meshID = mesh-fleet` ) via Istio **multi-primary, multi-network**.

#### How they're joined

- **Shared root CA** → workloads in EKS & GKE trust each other's mTLS.
- **A primary istiod per cluster** — no control-plane SPOF.
- **East-west gateways** (LB) carry cross-cluster mTLS traffic.
- **Remote secrets** exchanged → each istiod discovers the other's endpoints.
- **Network labels** `topology.istio.io/network` (`network-aws` / `network-gcp`) — the key to cross-network discovery.

#### Global service & LB

`hello.demo.svc.cluster.local` runs in both clusters. A `DestinationRule` sets an explicit **50/50 cross-region split** with `outlierDetection` so a failed cloud is ejected and traffic redistributes (active-active failover).

East-west gateways: EKS `...elb.amazonaws.com` · GKE `136.116.250.171`.

#### ✓ Verified active-active behaviour

From a client pod in **EKS**, 20 calls to the global service returned **aws: 12 / gcp: 8** — traffic from one cloud is served by **both** clouds through the joined mesh.

```
# from an EKS pod
for i in $(seq 20); do curl -s hello.demo.svc.cluster.local; done
hello from aws    x12
hello from gcp    x8
```

## 5 Tasks completed

| #  | TASK                                                             | STATUS           |
|----|------------------------------------------------------------------|------------------|
| 1  | Tooling (gh, terraform, aws, gcloud, az, kubectl, istioctl)      | ✓                |
| 2  | Cloud identities via OIDC (AWS role, GCP WIF)                    | ✓                |
| 3  | Remote state (S3 native locking)                                 | ✓                |
| 4  | CI/CD pipelines + gated production env                           | ✓                |
| 5  | Provision EKS (VPC, node group)                                  | ✓                |
| 6  | Provision GKE (Standard, private nodes)                          | ✓                |
| 7  | Local kubectl access to both clusters                            | ✓                |
| 8  | Istio multi-primary install on both                              | ✓                |
| 9  | Join clusters (CA + east-west + remote secrets + network labels) | ✓                |
| 10 | Global app + active-active 50/50 load balancing                  | ✓ verified       |
| 11 | Observability (Prometheus/Grafana on EKS)                        | ✓                |
| 12 | Cross-cloud VPN module (AWS↔GCP HA VPN + BGP)                    | built, gated off |
| 13 | CockroachDB central DB (multi-region over VPN)                   | manifests only   |
| 14 | Control-plane container (agentic reconciler + dashboard)         | pending          |
| 15 | Global DNS (north-south LB)                                      | module built     |

### Notable blockers resolved

| SYMPTOM                                                  | FIX                                                        |
|----------------------------------------------------------|------------------------------------------------------------|
| EKS rejected K8s 1.29                                    | bump to 1.32                                               |
| GKE Autopilot can't host self-managed Istio / DaemonSets | switch to GKE Standard                                     |
| Node pool rejected <code>GKE_METADATA</code>             | enable Workload Identity on cluster                        |
| istiod won't schedule on medium nodes                    | 4 vCPU/16GB nodes (m5.xlarge / e2-standard-4)              |
| GCP free-trial IP quota (4) exceeded                     | private nodes + Cloud NAT                                  |
| Terraform state drift (partial applies)                  | fresh plan reconciled                                      |
| EKS sidecar-injector webhook timeout                     | open node SG :15017 / :15012                               |
| DestinationRule distribute+failover rejected             | distribute + outlierDetection                              |
| Cross-cluster traffic stayed local                       | add <code>topology.istio.io/network</code> namespace label |

## 6 Deployed vs pending · operate · costs

### ✓ Live now

EKS + GKE; Istio multi-primary mesh (joined); global app with active-active cross-cloud LB; observability on EKS; full CI/CD with OIDC + gated approvals.

### ⌚ Pending

Cross-cloud VPN (gated off); central **CockroachDB** (needs VPN first); control-plane container; global DNS (north-south).

### Operate

```
# access
aws eks update-kubeconfig --name bullion-eks --region us-east-1 --alias eks
```

```
gcloud container clusters get-credentials bullion-gke --zone us-central1-a --project project-f38bfcd6-70ff-4a8d-9cd
# deploy: PR → plan, merge → gated apply ; teardown: Actions → deploy → action=destroy
```

**Cost & security:** both clusters + LoadBalancers bill while running — run the `destroy` workflow when idle. Rotate the bootstrap AWS key (shared in plaintext during setup). The `Olisa` user holds AdministratorAccess (scope down beyond testing); GKE control-plane endpoint is public for CI/local access.

Bullion Infrastructure · active-active multi-cloud Kubernetes · repo Olisaemeka111/bullion-infrastructure · report generated 2026-06-23. — End of report.